

```
Open [icon]
1 Name - Sanskruti Gorle
2 PRN - 24070521025
3 Batch - A1
4
5
6 %{
7 #include <stdio.h>
8
9 int stringToNumber(char s[])
10 {
11     int n = 0;
12     int i = 0;
13
14     while(s[i] >= '0' && s[i] <= '9')
15     {
16         n = n * 10 + (s[i] - '0');
17         i++;
18     }
19
20     return n;
21 }
22
23 void numberToHex(int n)
24 {
25     char hex[] = "0123456789ABCDEF";
26     char result[20];
27     int i = 0;
28
29     if(n == 0)
30     {
31         putchar('0');
32         return;
33     }
34
35     while(n > 0)
36     {
37         result[i] = hex[n % 16];
38         n = n / 16;
39         i++;
40     }
41
42     while(i > 0)
43     {
44         i--;
45         putchar(result[i]);
46     }
47 }
48 %}
49
50 %%
51 [0-9]+ {
52     int n;
```

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*dect
24 {
25     char hex[] = "0123456789ABCDEF";
26     char result[20];
27     int i = 0;
28
29     if(n == 0)
30     {
31         putchar('0');
32         return;
33     }
34
35     while(n > 0)
36     {
37         result[i] = hex[n % 16];
38         n = n / 16;
39         i++;
40     }
41
42     while(i > 0)
43     {
44         i--;
45         putchar(result[i]);
46     }
47 }
48 %}
49
50 %%
51 [0-9]+ {
52     int n;
53
54     n = stringToNumber(yytext);
55
56     printf("Decimal = %d\n", n);
57     printf("Hexadecimal = ");
58
59     numberToHex(n);
60
61     printf("\n");
62 }
63 %%
64
65 int main()
66 {
67     printf("Enter decimal number: ");
68     yylex();
69
70     return 0;
71 }
72
73 int yywrap()
74 {
75     return 1;
76 }
```

```
sanskruti@LAPTOP-ICU7078D:~$ gedit dectohex.l
sanskruti@LAPTOP-ICU7078D:~$ flex dectohex.l
sanskruti@LAPTOP-ICU7078D:~$ gcc lex.yy.c
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter decimal number: 55
Decimal = 55
Hexadecimal = 37

36
Decimal = 36
Hexadecimal = 24

39
Decimal = 39
Hexadecimal = 27

25
Decimal = 25
Hexadecimal = 19

789
Decimal = 789
Hexadecimal = 315

10
Decimal = 10
Hexadecimal = A
Sanskriti Gorle
```

```
Open [ + ]
1 Name - Sanskruti Gorle
2 PRN - 24070521025
3 batch A1
4
5 %
6 #include <stdio.h>
7
8 int hexToNumber(char s[])
9 {
10     int n = 0;
11     int i = 0;
12     int value;
13
14     while(s[i] != '\0')
15     {
16         if(s[i] >= '0' && s[i] <= '9')
17         {
18             value = s[i] - '0';
19         }
20         else if(s[i] >= 'A' && s[i] <= 'F')
21         {
22             value = s[i] - 'A' + 10;
23         }
24         else if(s[i] >= 'a' && s[i] <= 'f')
25         {
26             value = s[i] - 'a' + 10;
27         }
28
29         n = n * 16 + value;
30         i++;
31     }
32
33     return n;
34 }
35 %
36
37 %
38 [0-9A-Fa-f]+ {
39     int n;
40
41     n = hexToNumber(yytext);
42
43     printf("Hexadecimal = %s\n", yytext);
44     printf("Decimal = %d\n", n);
45 }
46 %
47
48 int main()
49 {
50     printf("Enter hexadecimal number: ");
51     yylex();
52
53     return 0;
54 }
```

Bracket match found on line: 35

```
Open [ + ]
7
8 int hexToNumber(char s[])
9 {
10     int n = 0;
11     int i = 0;
12     int value;
13
14     while(s[i] != '\0')
15     {
16         if(s[i] >= '0' && s[i] <= '9')
17         {
18             value = s[i] - '0';
19         }
20         else if(s[i] >= 'A' && s[i] <= 'F')
21         {
22             value = s[i] - 'A' + 10;
23         }
24         else if(s[i] >= 'a' && s[i] <= 'f')
25         {
26             value = s[i] - 'a' + 10;
27         }
28
29         n = n * 16 + value;
30         i++;
31     }
32
33     return n;
34 }
35 %
36
37 %
38 [0-9A-Fa-f]+ {
39     int n;
40
41     n = hexToNumber(yytext);
42
43     printf("Hexadecimal = %s\n", yytext);
44     printf("Decimal = %d\n", n);
45 }
46 %
47
48 int main()
49 {
50     printf("Enter hexadecimal number: ");
51     yylex();
52
53     return 0;
54 }
55
56 int yywrap()
57 {
58     return 1;
59 }
```

```
sanskruti@LAPTOP-ICU7078D:~$ gedit hextodec.l
sanskruti@LAPTOP-ICU7078D:~$ flex hextodec.l
sanskruti@LAPTOP-ICU7078D:~$ gcc lex.yy.c
sanskruti@LAPTOP-ICU7078D:~$ ./a.out
Enter hexadecimal number: 552
Hexadecimal = 552
Decimal = 1362

AAA
Hexadecimal = AAA
Decimal = 2730

ABC
Hexadecimal = ABC
Decimal = 2748

56F
Hexadecimal = 56F
Decimal = 1391

7B
Hexadecimal = 7B
Decimal = 123
Sanskruti Gorle
```